A Literature Review of Dataset Reconstruction from Released Models and Statistics

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1 Introduction

This review is about *dataset reconstruction attacks* on tabular data. In such an attack, an attacker looks at something that was published, for example a decision tree with the number of examples of each class in every node, a random forest, or a set of statistical tables, and tries to recover the records that were used to make it. Two goals conflict here. On the one hand, interpretable models are recommended for high-stakes decisions (Rudin, 2019), and transparency requirements are a reason to publish trained models (Ferry et al., 2024b, Impact Statement; AI Act Art. 13 in Gorgé et al., 2026, Sec. I). On the other hand, a tree that is published with its leaf counts can be seen as a small table of statistics, and census tables have already been turned back into records of individual people for an entire country (Abowd et al., 2025).

The starting paper of this review, Ferry et al. (2024a), measures how much can be reconstructed from a single decision tree or rule list by computing the entropy over all datasets that fit the published model. Starting from this paper, seven peer-reviewed papers from 2005 to 2026 are compared to answer four research questions (RQs): how the sources of leakage and the attacks changed over time (**RQ1**), how the attacks are evaluated (**RQ2**), which defences were tested (**RQ3**) and what is still missing (**RQ4**). The main finding is that most attacks return one single reconstructed dataset without saying how likely each value is to be correct, which leads to the research niche proposed at the end.

2 Method

Search. The starting paper is called P0 and the seven reviewed papers P1 to P7 (Table 2). The reference list of P0 led to P2. The other papers were found with web searches, each on a different part of the topic. Table 1 lists the web searches that led to the included papers (W1 to W5) and eight keyword searches on DBLP (D1 to D8). While reading, it became clear that the papers also cite each other: P4 and P5 cite P0, P7 cites P4 and P6 cites P3. A systematic search for all papers citing P0 was not done.

Inclusion and exclusion. Only peer-reviewed papers with a full text available were included. The papers also had to be about tabular data. Each paper had to describe either an attack that rebuilds records (or one full column) from something that was published, such as a model, statistics or microdata, or a defence against such an attack. Papers about reconstructing images or text through gradients or federated learning were left out. Papers that only study membership inference were also left out, just like surveys and preprints that were not peer-reviewed. There was no limit on the publication year. After going through the titles and short descriptions, the remaining papers were read in full and seven of them were kept. The paper by Gambs et al. (2012), is only used as background, because the starting paper is a generalisation of it. P1 was added on purpose, because

Table 1: Search queries. Web searches: number of top results checked and the papers kept from them. DBLP: total number of hits. No filters were used.

Web search				DBLP					
W1	dataset reconstruction attack interpretable models decision trees	9	P0	D1	dataset reconstruction	306	D5	reconstruction census	11
W2	training set reconstruction attack random forest	10	P4, P5	D2	reconstruction attack	364	D6	reconstruction random forest	12
W3	census reconstruction attack published statistics	10	P3, P6	D3	training set reconstruction	5	D7	Rashomon privacy	0
W4	Rashomon set privacy reconstruction	8	P7	D4	reconstruction interpretable	39	D8	sensitive attributes reconstruction	2
W5	optimal k-anonymization	9	P1						

Table 2: The starting paper (P0) and the seven reviewed papers. †Training-set size inferred from the reported split. ‡Stronger baselines only in an appendix setting with $N-1$ known records.

Paper	Released artefact	Method (solver)	Output	Largest n	External baseline	Defence tested
P0 Ferry et al. (2024a)	tree or rule list with counts	counting, entropy (none)	probabilistic (possible worlds)	≈5,771†	none	none
P1 Bayardo & Agrawal (2005)	k-anonymised microdata	pruned set-enumeration search (custom)	optimal anonymisation	30,162	incomplete optimisers	k-anonymity, not attacked
P2 Ferry et al. (2023)	fair-model outputs, tolerance	IP/CP (CP Optimizer)	one sensitive column	≈45,308†	ML attribute inference	hiding fairness information
P3 Dick et al. (2023)	exact tables, marginals	randomised relaxation, 100 runs	ranked rows, uncalibrated	≥68,369 (ACS state)	geographic, holdout sampling	none
P4 Ferry et al. (2024b)	random forest with counts	CP (CP-SAT); MILP (Gurobi)	one dataset	1,500	uniform random	bagging
P5 Gorgé et al. (2026)	ε-DP random forest	CP, noise likelihood (CP-SAT)	one most likely dataset	500	uniform random‡	pure ε-DP
P6 Abowd et al. (2025)	34 census table sets	ILP (Gurobi)	one dataset, provable bounds	308.7M	modal, proportional guessers	DP, swapping
P7 Hsu et al. (2026)	1–200 Rashomon trees	P4's CP attack	scalar error	100	uniform random	disclosure size

it is a paper on the defence side (optimal k-anonymisation). In the rest of this review, counts such as 2/7 are always taken over the seven reviewed papers, and the starting paper is discussed separately.

3 Synthesis of findings

3.1 Leakage sources: from anonymised tables to trained models (RQ1)

The oldest paper sees privacy as a problem of publishing microdata. Bayardo & Agrawal (2005, Sec. 7.1) compute provably optimal k-anonymisations (Sweeney, 2002) of 30,162 Adult records. Their guarantee is stated for identification "through linking attacks alone" (Bayardo & Agrawal, 2005, Sec. 1), and no discussion of statistics or models as a source of leakage was found in the paper. A second line of work goes back to Dinur & Nissim (2003), who reconstructed the private bits of a database from answers to queries. This result is cited by 5/7 papers and by the starting paper. For attacks that use the structure of a model, the starting paper builds on and generalises Gambs et al. (2012) (Ferry et al., 2024a, Sec. I).

The papers from 2023 on can be grouped into two tracks. The first track looks at published statistics. Dick et al. (2023, Sec. 2.3)[1] rebuild census rows from exact tables and marginals, and conclude that aggregating data is not enough to protect privacy. Abowd et al. (2025, Abstract; Sec. 1.1, 2.2, 2.4) do the same for all 308,745,538 persons of the 2010 Census, using integer programming on 34 of the 180 table sets. The second track looks at trained models. Ferry et al. (2023, Sec. III) use the predictions of a fair model and its public fairness tolerance as constraints on a sensitive column. The starting paper turns a single tree or rule list into a set of possible worlds, which are all the datasets that fit the model (Ferry et al., 2024a, Sec. III). After that, several model papers take up a problem that an earlier paper left open. The starting paper leaves ensembles out of scope (Ferry et al., 2024a, Sec. VI), and Ferry et al. (2024b, Sec. 3, 6) then combine all trees of a random forest using constraint programming (CP). They leave differential privacy (DP; Dwork et al., 2006) for future work (Ferry et al., 2024b, Sec. 8), and Gorgé et al. (2026, Sec. III) extend their attack to the noisy counts of DP forests. The starting paper also suggests combining the knowledge of several probabilistic reconstructions (Ferry et al., 2024a, Sec. VI), and Hsu et al. (2026, Sec. 6.2) publish up to 200 near-optimal trees from one Rashomon set, although they do not cite the starting paper. These steps made the attacks larger or more realistic, but none of P4, P5 and P7 uses the entropy measure of the starting paper.

Comparing the two tracks suggests that they attack the same kind of object: counts that act as linear constraints on unknown records (Abowd et al., 2025, Sec. 2.4; Ferry et al., 2024b, Sec. 6). Still, none of the reviewed papers on one track cites a reviewed paper on the other track, and only 2/7 papers cite the starting paper (Ferry et al., 2024b; Gorgé et al., 2026). In the papers that change how much is published, publishing more usually leaks more. More and deeper trees lower the error of Ferry et al. (2024b, Sec. 7.2), more Rashomon models mostly lower the error of Hsu et al. (2026, Fig. 9), and two extra tables raise the share of census blocks with a provably unique reconstruction from 65% to 70% (Abowd et al., 2025, Sec. 2.6). DP is the exception. For a fixed ε, forests with more than about 5–10 trees often become *less* vulnerable (Gorgé et al., 2026, Result 2).

3.2 Attack formulations and output types (RQ1)

Three types of attack can be found. Counting combined with entropy is only used by the starting paper (0/7). Exact CP or integer programming is used by 5/7 papers (Ferry et al., 2023; 2024b; Gorgé et al., 2026; Abowd et al., 2025; Hsu et al., 2026). Randomised optimisation with ranking is used by 1/7: Dick et al. (2023, Sec. 1.2, App. A) run a relaxed optimiser 100 times and rank the rows by how often they come back. The papers differ even more in the type of output. In 4/7 papers the output is one single reconstruction (Ferry et al., 2023; 2024b; Gorgé et al., 2026; Hsu et al., 2026). Abowd et al. (2025, Sec. 2.6) also give a provable bound on how much two reconstructions that fit the data can differ. Dick et al. (2023, Sec. 3) rank the rows, but say themselves that this ranking is not calibrated. **None of the seven papers (0/7) outputs a probability for each reconstructed value**. Only the starting paper does this. Some papers mention uncertainty that they do not measure. For example, Ferry et al. (2024b, App. E) mention free attributes that can take any value without changing the likelihood, so a correct guess there may just be luck.

The papers differ on two points. *Is it useful to collect more than one solution?* For Dick et al. (2023, Introduction), ranking many random reconstructions is the main new idea. Abowd et al. (2025, Sec. 2.6), on the other hand, see "no real advantage to collecting additional solutions", because their bound is already small. Comparing the two suggests that each view fits its own setting. The census with 34 tables has little solution variability, while bagged forests still leave about 5–10% of

[1]Section, figure and appendix numbers for Dick et al. (2023) refer to its arXiv version (arXiv:2211.03128).

the data unrecovered (Ferry et al., 2024b, App. E). There, looking at many different datasets that all fit the forest, instead of only the one the solver returns, could show which values are reliable and which are only guesses. *Single trees or forests?* The starting paper sees ensembles as a challenge for its independence assumption (Ferry et al., 2024a, Sec. IV-A). Ferry et al. (2024b, Sec. 7) show that without bagging, enough deep trees bring the error down to zero, but no comparison with the starting paper was found in P4, even though P4 also tests forests with a single tree.

3.3 Evaluation protocols (RQ2)

Adult is used by 5/7 papers and by the starting paper, and COMPAS and Default Credit by 3/7. Attacks that rebuild full records from published ensembles are only tested on small data: at most 1,500 records (Ferry et al., 2024b, Table 6), 500 with DP (Gorgé et al., 2026, Table VII) and 100 for Rashomon sets (Hsu et al., 2026, App. D.3). P4, P5 and P7 also turn every feature into a binary one (3/7, plus the starting paper). The papers measure success in different ways (remaining entropy, wrong binary cells, exact row matches, re-identification precision or the accuracy on one column), which makes the results hard to compare. Hsu et al. (2026) report the error of P4's attack without defining it, and none of the papers was found to measure calibration (0/7).

Baselines differ a lot. In their main experiments, 3/7 papers only compare with uniform random guessing (Ferry et al., 2024b; Gorgé et al., 2026; Hsu et al., 2026), which Hsu et al. (2026, App. D.3) call "a very conservative attack". 3/7 papers use baselines that know the data distribution: attribute inference trained on data from the same distribution (Ferry et al., 2023, Sec. IV-A), geographic and holdout sampling (Dick et al., 2023, Sec. 2.2), or guessers that pick the most common value or draw values in proportion to how common they are (Abowd et al., 2025, Sec. 3.7). The choice of baseline can change the conclusion. Over the whole country, census reconstruction re-identifies people *worse* than simply guessing the most common value (41.2% vs. 45.9% precision with commercial data; Abowd et al., 2025, Table 5). The attack does much better for nonmodal persons, whose race and ethnicity differ from the most common one in their block. In blocks of 1–99 persons, the precision is above 50% for about 23.8 million of them (Abowd et al., 2025, Sec. 3.9). Among attacks on tree models, only P5 was found to use such a stronger baseline (Gorgé et al., 2026, Result 4).

3.4 Defences and the privacy–utility trade-off (RQ3)

Only 2/7 papers test a formal defence. Gorgé et al. (2026, Sec. IV) attack pure ε-DP forests with $\varepsilon \in \{0.1, 1, 5, 10, 20, 30\}$, and Abowd et al. (2025, Sec. 4) compare the 2020 census DP system with 5% and 50% swapping. Three other papers look at other protections: bagging only partly protects the data (Ferry et al., 2024b, Sec. 7.2), hidden fairness information can be estimated by the attacker (Ferry et al., 2023, Sec. V-B), and Hsu et al. (2026, App. E.8) limit how many models are published. Earlier, Ferry et al. (2023, Sec. V-A) argued, without testing it, that DP cannot stop their attack, and Dick et al. (2023, Introduction) called imprecise statistics the only defence. The later DP experiments can be read as partly supporting both views. "Only DP RFs with trivial predictive performance" give the attacker no advantage over random guessing (Gorgé et al., 2026, Result 3), and at $\varepsilon = 1$, 22.0% of the COMPAS records are still recovered exactly (10 trees of depth 5; Gorgé et al., 2026, Table III). For the most exposed census group (nonmodal persons who are unique on sex and age group in a provably reconstructed block), DP lowers the precision to 21.6–21.8%, close to the proportional guesser. With 50% swapping, it is still at least 76.4% (Abowd et al., 2025, Table 6). Comparing P5 and P6 suggests that, of the tested defences, only DP brings the attacks close to a simple guessing baseline (random guessing in P5, a proportional guesser in P6), and in P5 this costs predictive accuracy.

4 Gaps and open problems (RQ4)

G1. Missing (0/7): calibrated probabilistic reconstruction for more than one model. The authors point out several open problems themselves. The starting paper leaves ensembles open (Ferry et al., 2024a, Sec. VI), its method cannot directly deal with DP noise (Gorgé et al., 2026, Result 3), and Dick et al. (2023, Sec. 3) call calibration "a concrete technical problem". Comparing the papers suggests one more point: the CP models of P4 and P5 encode a whole set of datasets that fit the published model, but return only one of them.

G2. Weakly explored (1/7): publishing several models. Only Hsu et al. (2026) publish several trained models, P7 was only found through a web search, and the DBLP search for `Rashomon privacy` (D7) gave no results. The starting paper suggests combining several probabilistic reconstructions (Ferry et al., 2024a, Sec. VI), and Hsu et al. (2026, Sec. 7) call for work on disclosure policies. No statement was found in P7 on what information the attack gets about each tree.

G3. Evaluation gap (3/7 only use random guessing as a baseline, 0/7 measure calibration). Apart from P5 (Result 4, App. G), no attack on tree models was found to use a baseline that knows the marginals or a holdout set. A stronger baseline can turn the conclusion around (Abowd et al., 2025, Table 5).

G4. Missing (0/7): reconstruction from DP models that also shows its uncertainty. The starting paper says that noisy DP counts would cause errors in its reconstruction (Ferry et al., 2024a, App. C), and Gorgé et al. (2026, Sec. VI) leave the use of prior knowledge about the data for future work. In addition, P5 does not test any ε between 0.1 and 1.

G5. Data and model types. Attacks on published tree ensembles are only tested on binarised data, with at most 1,500 records and no missing values. Rule lists are attacked in experiments only by the starting paper (0/7), and no attack on boosted trees was found (0/7). Ferry et al. (2024b, Sec. 8) name other types of ensembles as future work.

G6. Reproducibility and harm. Code is available for 6/7 papers, but the census re-identification uses confidential data, and no solver time limit was found in P6 and P7.

5 Conclusion: research niche

Based on gaps G1 to G3, the proposed niche is **measuring how uncertain a reconstruction is when several interpretable models from one Rashomon set are published, and comparing the attack with baselines that know the data distribution**.

Research question: Take sparse decision trees from one Rashomon set, trained on $n = 100$ binarised records and published with their leaf counts. How fast does the normalised entropy Dist_G of the starting paper go down as the number m of published trees grows, are the probabilities for each reconstructed value calibrated, and how much of the leakage goes beyond a baseline based on holdout marginals? This can be tested with the public code of P4 and P7, using the setup of Hsu et al. (2026, App. D.3).

Why now and why important: transparency requirements encourage publishing models, and Hsu et al. (2026, Sec. 7) suggest publishing representative models, but their tree experiments measure leakage with the average error of one reconstruction. It can be argued that a rule on how many models can be published should also know which records become certain. Current attacks on tree ensembles find one dataset that fits a published model. The next question is how many other datasets also fit, and for which people no uncertainty is left.

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